

changing the controller program. Industrial developer boards reduce the bill-of-material and space requirements as these can handle SMD components.

Pallav Aggarwal, engineering manager, Toradex systems India, says that factors like longevity, warranty and flexibility make industrial boards a preferred option compared to normal boards. He adds that industrial temperatures may range from -40 degrees to +80 degrees centigrade from industry to industry.

Interestingly, existing industrial developer boards are all geared up for innovation in the year 2017. We bring to you the latest trends in industrial boards used in some of the most dynamic and fast-growing industries.

Moving on from silicon to gallium nitride

Earlier, industrial developer boards used silicon technology field-effect transistors (FETs) for power electronics applications in automotive electrical and electronic systems, and commercial, industrial or residential utilities. However, these have now graduated to gallium nitride as it can support higher-frequency-driven systems like boost LED drivers of higher wattages and systems utilising higher voltage output (up to 1kV). GaN FETs are commonly used in servers and LED drivers. The design of these FETs improves thermal dissipation, providing much better efficiency as well as high-speed switching.

Equipped with RF transmitter

One of the major changes in industrial boards has happened in the robotics segment. Keeping up with the latest trends, industrial developer boards have now been revamped with both hardware and software upgrades.

Earlier, robots were manu-

ally power consumption by them. To address this issue, the design of industrial developer boards used in these robots has been upgraded.

According to Akshat Garg, analogue application engineer from Texas Instruments, manufacturers have now equipped these boards with an RF-based detector that uses antenna instead of optics. In order to make the board compact, the antennae are embedded in the system. This has not only condensed the board but also reduced the processing time.

Besides, the industrial developer board software has been completely transformed from normal image processing to focus more on XYZ coordinates in order to enable robots to easily identify objects in 3D space.

These developer boards are now being used by designers in heavy automation segments like conveyor belts, pick-and-place machinery, oil drilling rigs (which go very deep and impossible to operate manually), and placement of gems or diamonds on jewellery.

Better connectivity

One of the major changes in industrial developer board design is the transition from Bluetooth 4.2 to the latest version Bluetooth 5.0.

Earlier, the lighting application (primarily in sectors such as lighting, automotive and aviation) used Bluetooth 4.2, which did not have mesh facility in the developer board. Thus, the connectivity between different systems was limited. Nevertheless, with advancement to Bluetooth 5.0, systems are now interconnected with each other and can be accessed by a single remote. This has further enhanced the exchange of data exponentially as the speed of data transfer has increased and the time reduced.

This facility is embedded in the microcontroller on the development board itself.

Multipurpose boards

Prahlad Supeda, senior application engineer, Texas Instruments, adds that till recently, customers were provided with different industrial boards for different applications. However, this is changing as the demand for a compact and complete industrial board is rapidly growing.

For instance, aviation and automobile manufacturers used separate developer boards for dashboards incorporating features for telematics control, radar, body electronics and lighting. From the dashboard of the vehicle to the cockpit of the plane, all have a control panel that consists of speedometer, accelerometer, fuel indicator and many such instruments. These were all controlled through different industrial developer boards.

But now, automobile manufacturers are producing cars that are usually well equipped with Bluetooth, Wi-Fi, indicators, odometer, fuel indicator, theft alarm and many such features. And all this is being done using a single board and a controller that is equipped with all these features.

Similarly, infotainment system, cluster, automotive and advanced driver-assistance systems are integrated on a single MCU. This reduces the wire harnessing, simplifying the anatomy of cars while making these safer as well as far more efficient.

Last but not the least

Industrial developer boards are constantly evolving. However, be it a replacement for components upgrade or protocol upgrade, manufacturers and customers always go for a combination of low cost and best in technology. **EFY**